

INTRODUCTION TO GIT GENETICS ABNORMALITY

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Genetic Abnormalities in The Gastrointestinal Tract

Genetic abnormalities in the gastrointestinal (GI) tract can manifest in various forms, each with distinct implications for disease development and progression. Here's a more detailed look at the types of genetic abnormalities affecting the GI tract:

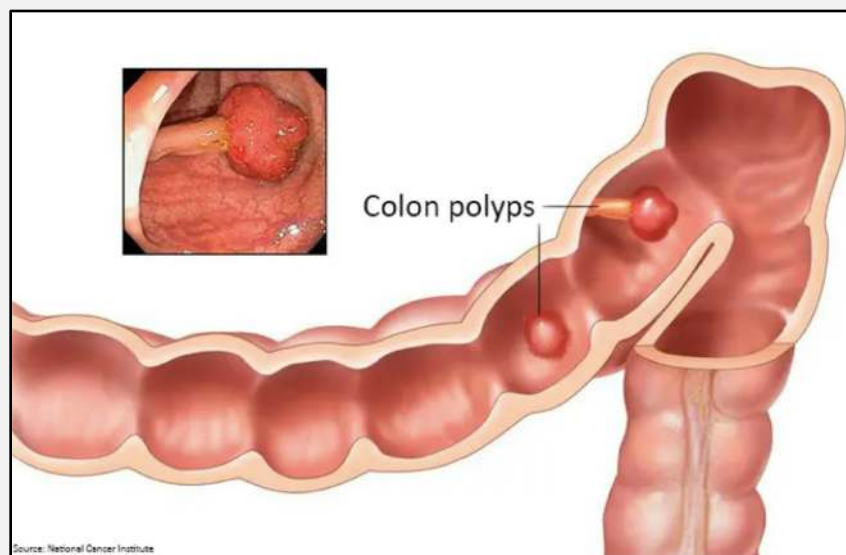
I. Single-Gene Disorders

Single-gene disorders are caused by mutations in a single gene and can have profound effects on the GI tract.

a. Familial Adenomatous Polyposis (FAP)

- **Gene Involved:** Adenomatous Polyposis Coli (APC) gene.
- **Mechanism:** The Adenomatous Polyposis Coli (APC) gene is a tumor suppressor gene that regulates cell growth. Mutations lead to the loss of its function, resulting in the formation of hundreds to thousands of adenomatous polyps in the colon and rectum during adolescence or early adulthood.
- **Clinical Implications:** If left untreated, there is nearly a 100% risk of developing colorectal cancer. FAP is also associated with extracolonic manifestations such as duodenal polyps and desmoid tumors.

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b. Hereditary Nonpolyposis Colorectal Cancer (HNPCC) / Lynch Syndrome

- **Genes Involved:** Mismatch repair (MMR) genes, primarily MLH1, MSH2, MSH6, and PMS2.
- **Mechanism:** Mutations in MMR genes lead to microsatellite instability and an increased mutation rate in cells, which predisposes individuals to colorectal and other cancers, including endometrial, gastric, and ovarian cancers.
- **Clinical Implications:** Lynch syndrome accounts for 3-5% of all colorectal cancers. Patients have a lifetime colorectal cancer risk of up to 80%, and regular screening is crucial.

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c. Peutz-Jeghers Syndrome (PJS)

- **Gene Involved:** Serine/Threonine Kinase (STK11) also known as Liver Kinase B1 (LKB1).
- **Mechanism:** STK11 is a tumor suppressor gene, and its mutation results in the development of hamartomatous polyps throughout the GI tract, especially in the small intestine.
- **Clinical Implications:** There is an increased risk of GI cancers (including colorectal, gastric, and pancreatic) as well as non-GI cancers (such as breast and ovarian cancer). The characteristic feature is mucocutaneous pigmentation (dark spots on the lips and mouth).

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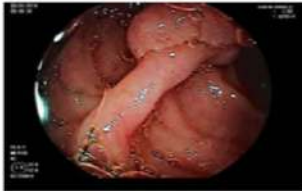
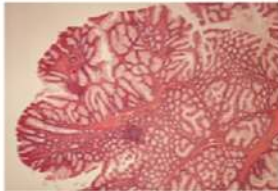
Syndrome	Phenotype	Endoscopy	Histology	Gene
Peutz-Jeghers syndrome	Muco-cutaneous pigmentation Gastro-intestinal hamartomatous polyps Colorectal, pancreatic, gynecological cancers			<i>STK11</i>

Fig. 1. Genetic hamartomatous polyposis syndromes: clinical, endoscopic and histological features.

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Hamartomatous Tumors in the Gastrointestinal Tract

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Hamartomatous (Peutz Jeghers) polyp of the colon in an adult

Hamartomatous polyps of the gastrointestinal tract can occur sporadically.



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d. Juvenile Polyposis Syndrome (JPS)

- **Genes Involved:** Bone Morphogenetic Protein Receptor Type IA (BMPRI A) and Small Mothers Against Decapentaplegic (SMAD4).
- **Mechanism:** Mutations in these genes disrupt the transforming growth factor-beta (TGF- β) signaling pathway, leading to the formation of juvenile polyps, which are hamartomatous and primarily occur in the colon.
- **Clinical Implications:** Patients have an increased risk of colorectal cancer and, in some cases, gastric cancer.

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2. Chromosomal Abnormalities

Chromosomal abnormalities involve changes in the structure or number of chromosomes, which can lead to various GI anomalies.

a. Down Syndrome (Trisomy 21)

- **Mechanism:** Presence of an extra copy of chromosome 21 affects multiple systems, including the GI tract.

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- **Clinical Implications:** GI manifestations include:

1. Hirschsprung Disease: Absence of nerve cells in parts of the intestine, leading to severe constipation or obstruction.

2. Duodenal Atresia: Congenital closure of the duodenum, requiring surgical intervention.

3. Imperforate Anus: Absence of a normal anal opening.

4. Esophageal Atresia: The esophagus ends in a blind pouch rather than connecting to the stomach.

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b. Turner Syndrome (45, X)

- **Mechanism:** Partial or complete absence of one of the X chromosomes in females.

- **Clinical Implications:**

GI manifestations are less common but may include congenital malformations like pyloric stenosis or inflammatory bowel disease.

- Pyloric stenosis is a narrowing of the opening from the stomach to the first part of the small intestine.

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3. Complex Polygenic Disorders (Multifactorial disorder)

These involve the interaction of multiple genes, each contributing a small risk, combined with environmental factors.

a. Inflammatory Bowel Disease (IBD)

- **Genes Involved:** Nucleotide-binding oligomerization domain-containing protein 2 (NOD2) have been associated with Crohn's disease and ulcerative colitis.
- **Mechanism:** Genetic predispositions interact with environmental factors, leading to an abnormal immune response to gut microbiota.

- **Clinical Implications:**

1. Crohn's Disease: Can affect any part of the GI tract and is characterized by patchy inflammation, strictures, and fistulas.

2. Ulcerative Colitis: Primarily affects the colon and rectum, causing continuous inflammation and ulcers.

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b. Celiac Disease

- **Genes Involved:** HLA-DQ2 and HLA-DQ8.
- **Mechanism:** These HLA genes are involved in the immune system's response to gluten. In genetically susceptible individuals, ingestion of gluten leads to an immune-mediated destruction of the small intestinal villi.
- **Clinical Implications:** Symptoms include malabsorption, diarrhea, and anemia. Long-term untreated celiac disease can lead to an increased risk of small bowel lymphoma.

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4. Mitochondrial Disorders

Mitochondrial DNA mutations can lead to multi-systemic diseases with GI involvement.

a. Mitochondrial Neurogastrointestinal Encephalomyopathy (MNGIE)

- **Genes Involved:** Thymidine phosphorylase (TYMP) gene, which encodes thymidine phosphorylase.
- **Mechanism:** Deficiency in this enzyme leads to toxic levels of nucleosides, causing damage to the mitochondria in various tissues, including the GI tract.
- **Clinical Implications:** Symptoms include severe GI dysmotility, pseudo-obstruction, and malabsorption, along with neurological deficits.

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5. Epigenetic and Somatic Mutations

These involve non-inherited changes that can affect gene expression and contribute to GI diseases, particularly cancer.

a. Epigenetic Changes

- **Mechanism:** Abnormal DNA methylation or histone modification can silence tumor suppressor genes or activate oncogenes.
- **Clinical Implications:** Epigenetic alterations play a significant role in the development of colorectal cancer and other GI malignancies.

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b. Somatic Mutations

- **Mechanism:** Mutations that occur in individual cells after conception, such as those in the Kirsten rat sarcoma virus (KRAS) or Tumor protein P53 (TP53) genes in colon cancer.
- **Clinical Implications:** These mutations are not inherited but can drive the development of sporadic GI cancers.
- KRAS (Kirsten rat sarcoma virus) is a gene that **provides instructions for making a protein called K-Ras**, a part of the RAS/MAPK pathway.

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